

Do Old Age Security (OAS) Payments Increase Retirement Among Canadians?

Andrew Clark (#91744607)

Yushin Nam (#85058832)

ECON 398: Introduction to Causal Methods

April 19, 2026

1 Introduction

The Old Age Security (OAS) pension is an important public retirement benefit that provides financial support to Canadians 65 years and older as they transition to retirement. This federally funded program is intended to help Canadians retire, but the extent of its effects remains uncertain. In this paper, we use data from the 2021 Canadian Census to determine whether eligibility for OAS has a causal effect on labour force participation rates in Canada.

We implement a difference-in-differences (DD) approach to analyze this effect, comparing labour force participation rates between OAS-eligible and OAS-ineligible groups before and after age 65. Under the common trends assumption, this DD approach controls for time effects and helps isolate differences in labour force participation rates that naturally exist between the eligible and ineligible groups.

In addition, we control for variables such as gender, marital status, and education in our regression model. This allows us to better estimate the interaction effect between eligibility status and age eligibility, which is our main parameter of interest. Estimating this effect can provide valuable insight into whether the OAS pension causally affects labour force participation rates. Our findings can help answer broader questions about how public pension programs influence retirement decisions among older Canadians.

2 Background

The Old Age Security (OAS) pension is a federally funded public retirement benefit available to eligible Canadians 65 years of age and older who meet residency requirements. Specifically, individuals must have resided in Canada for at least 10 years after turning 18 if they currently live in Canada, or at least 20 years after turning 18 if they currently reside outside of Canada. OAS is intended for middle- to low-income Canadians 65 years of age and older, with the option to defer payments in exchange for higher annual benefits up to age 75. A “recovery tax” is applied to individuals with annual incomes above approximately \$90,000, and OAS payments are fully taxed

back for individuals earning above approximately \$150,000 annually.

Furthermore, OAS differs from the Canada Pension Plan (CPP), as OAS payments are not based on lifetime earnings. Instead, its primary purpose is to provide a basic level of income support in retirement for eligible Canadians.

With this understanding of OAS, we note that non-labour income increases at retirement age for OAS-eligible Canadians, which should increase the likelihood of retirement according to a labour-leisure trade-off model from intermediate microeconomics. However, the magnitude of this effect is an empirical question that can be addressed using a difference-in-differences (DD) framework to isolate the effect of OAS eligibility on labour force participation.

3 Data

We are using data from the 2021 Canadian Census Public Use Microdata File (PUMF) on individuals. This dataset is one of the richest available on Canadian individuals, with 980,868 responses, representing 2.7% of the Canadian population, which was estimated to be 36,328,477 people. The 2021 PUMF contains 126 variables, with 95 of them coming from the individual universe, and 31 coming from the family, household, and dwelling universe. The reference date for questions in the census is May 11, 2021.

To construct an analysis sample from the PUMF, we focused on four variables, age (AGEGRP), immigration status (IMMSTAT), labour force status (LFACT), and age of immigration (AGEIMM). We are only including individuals from two age groups, “60 to 64 years” (code 16) and “65-69 years” (code 17). We excluded other ages because we only want to focus on the local effect of OAS eligibility, which can kick in at age 65 at the earliest. In our DD model, we are considering these two age groups as our two different time periods. We define $T_i = 0$ to be “60 to 64 years”, and $T_i = 1$ to be “65 to 69 years”.

Next, we want to exclude non-permanent residents and individuals who did not report an immigration status. We can use the immigration status variable (IMMSTAT) and only take responses

for “Non-immigrants” (code 1) and “Immigrants” (code 2). We exclude non-permanent residents since they are fundamentally different from non-immigrants and immigrants, and they can never be eligible for OAS.

We do not want to include individuals who are not in the labour force and have never worked since they never make a meaningful retirement decision. Thus, their inclusion would artificially lower the labour force participation rate and cause bias in our estimates. To accomplish this, we use the labour force status variable (LFACT) and exclude individuals who responded “Not in the labour force - Never worked” (code 14) or those who did not report a labour force status (code 88). We also note that there is a response “Not applicable” (code 15), but this applies to individuals under 15 years old who are already excluded from our analysis sample through our aforementioned usage of the age variable.

Additionally, we must exclude immigrants who did not report an age of immigration since we do not know whether they are OAS eligible or not. We use both the immigration status variable (IMMSTAT) from before and the age of immigration variable (AGEIMM) variable to exclude any individuals who selected “Immigrants” (code 2) to IMMSTAT and “Not available” (code 88) to AGEIMM.

Lastly, we need to exclude any individuals who did not respond to a variable that we are using as a covariate. Our covariates are listed below and their justifications include specifics on which variables have potential “Not available” responses.

With these restrictions in place, we have a good analysis sample to work with that allows us to focus on finding a causal effect for OAS eligibility on labour force participation. Our next step is to assign each individual from our analysis sample to either the OAS eligible ($G_i = 1$) or OAS ineligible ($G_i = 0$) group. The eligible group, $G_i = 1$, is composed of individuals who are eligible for OAS pension payments, or will be once they turn 65. It is important to note that we assign individuals who meet the OAS residency requirements but are not yet 65 years old to $G_i = 1$. Here, we are assuming that these individuals are comparable to true OAS eligible individuals who are over 65 and meet residency requirements, only at a different point in time. Moreover, residency

requirements state that individuals must have resided in Canada for at least 10 years after turning 18 if they currently live in Canada, or at least 20 years after turning 18 if they currently reside outside of Canada. We will make an assumption that all non-immigrants meet the residency requirements for OAS eligibility. While this may not hold in all cases, it is a reasonable assumption given that most Canadian-born individuals will have resided in Canada for a sufficient period.

We can filter for the eligible group, $G_i = 1$, using the variables immigration status (IMMSTAT) and age of immigration (AGEIMM). If an individual selected “Non-immigrants” for IMMSTAT, then we assign them $G_i = 1$ according to our aforementioned assumption. If an individual selected both “Immigrants” for IMMSTAT and any age group prior to and including 50-59 year group for AGEIMM, then we also assign them $G_i = 1$. Any individuals who do not fit either of these criteria are assigned $G_i = 0$.

With our analysis sample constructed and individuals properly assigned to a group, our next step is to choose and identify covariates using variables from the census data. We choose the following variables to act as covariates in our DD model:

- Gender of person (Gender)
- Marital Status (MarStH)
- Province or Territory of Current Residence (PR)
- Market Income (MrkInc)
- Household Size (HHSIZE)
- Housing Tenure (TENUR)
- Education (HDGREE)

Gender (Gender) is chosen because women generally retire earlier than men due to a number of reasons, such as increased family caregiving responsibilities or coordinating retirement dates with an older husband. For the marital status variable (MarStH), it is included because partnered

individuals can share household income and coordinate retirement decisions with their spouse, giving them greater financial flexibility to retire at or around 65.

Province or territory of current residence (PR) is chosen since different provinces and territories have different costs of living and tax rates which can affect retirement decisions. Market income (MrkInc) refers to the sum of employment income, investment income, private retirement income, and other income from market sources during the reference year of 2020 for an individual. The variable is continuous and we will divide each income value by 10,000 so that the estimated coefficients can be interpreted in thousands of dollars. We also need to remove any responses with the value “88,888,888” since this stands for “Not available”.

Household size (HHSIZE) is chosen because it may influence financial obligations and support systems, affecting an individual’s decision to retire. This variable takes values from “1 person” to “7 persons or more”, and we must exclude “Not available” responses. The housing tenure variable (TENUR) refers to whether the household owns or rents their private dwelling. Individuals who own their home may have higher financial stability than those who rent, which can impact retirement decisions. We must exclude “Not available” responses.

Our last covariate, education (HDGREE), is a very important one to consider because it is strongly correlated with lifetime earnings, job type, and retirement behaviour, influencing both labour force attachment and retirement decisions.

Lastly, our outcome variable representing labour force participation is determined by the labour force status variable (LFACT). We create a binary variable Y_i , where $Y_i = 0$ if an individual is in the labour force (i.e. codes 1 through 10), and $Y_i = 1$ if they are not in the labour force (i.e. codes 11 through 13). Using our constructed analysis sample and defined variables, we can estimate a regression model using a difference-in-difference approach.

4 Methodology

To estimate the effect of OAS eligibility on labour force participation, we implement a DD approach using data from the 2021 Canadian Census. This method compares differences in labour force participation between OAS eligible and OAS ineligible individuals before and after age 65. We estimate the following regression model:

$$Y_i = \beta_0 + \beta_1 G_i + \beta_2 T_i + \beta_3 (G_i \times T_i) + \sum_{k=1}^K \gamma_k X_{k,i} + \epsilon_i$$

Y_i represents the outcome, with $Y_i = 1$ representing individual i out of the labour force and $Y_i = 0$ representing individual i in the labour force. G_i represents which group individual i is in, with $G_i = 1$ representing the OAS eligible group, which includes individuals under 65 but who pass the OAS residency requirements. $G_i = 0$ represents the OAS ineligible group. T_i is a binary variable representing time, with $T_i = 0$ for the 60 to 64 year old individuals, and $T_i = 1$ for the 65 to 69 year old individuals.

$G_i \times T_i$ represents the interaction term between the group and time indicator, and captures the DD estimate. Each $X_{k,i}$ represents one of the covariates: gender, marital status, province or territory, market income, household size, housing tenure, and education. ϵ_i is the error term, capturing unobserved factors that affect labour force participation and are not included in the model.

We want to estimate β_3 , which is the DD effect. If $\beta_3 > 0$, then OAS eligibility increases the likelihood of not being in the labour force, meaning OAS eligibility increases the likelihood of retirement. If $\beta_3 < 0$, then OAS eligibility decreases the likelihood of being in the labour force. If $\beta_3 = 0$, then OAS eligibility has no effect on labour force participation.

However, to interpret β_3 causally, we rely on the common trends assumption, which requires that in the absence of OAS eligibility, labour force participation trends would have evolved similarly across the eligible and ineligible groups. This assumption allows us to attribute differences in changes over age groups to the effect of OAS eligibility rather than to underlying differences between groups or general age-related trends.

We also assume that no other concurrent policy or structural change differentially affects the eligible and ineligible groups at age 65. Additionally, the inclusion of covariates improves precision and controls for observable differences across individuals, although identification comes solely from variation in eligibility and age. Under these assumptions, the estimated coefficient β_3 can be interpreted as the causal effect of OAS eligibility on the probability of labour force participation.

5 Results

Table 1: DiD Regression Results

	<i>Dependent variable:</i>			
	Not in Labour Force (NILF)			Refined
	Baseline	Controls	Controls + factors	
	(1)	(2)	(3)	(4)
Eligible	0.138** (0.065)	0.136** (0.063)	0.138** (0.064)	0.144** (0.063)
Age 65 or older	0.404*** (0.090)	0.383*** (0.087)	0.386*** (0.087)	0.391*** (0.087)
Eligible x Age 65 or older	−0.096 (0.092)	−0.114 (0.089)	−0.120 (0.089)	−0.125 (0.089)
Demographic controls	No	Yes	Yes	Yes
Categorical controls as factors	No	No	Yes	Yes
Observations	2,568	2,568	2,568	2,568
R ²	0.099	0.168	0.179	0.174
Adjusted R ²	0.098	0.165	0.168	0.167

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 1 reports the results from our difference-in-differences (DD) estimation on the effect of OAS eligibility on labour force participation. We focus the table on the core difference-in-differences specification, which includes only the treatment indicator (eligibility), the time indicator (age 65+), and their interaction. Covariates are excluded from the main specification to

maintain clarity and focus on the identification of the DD effect, particularly given the large number of available controls. The inclusion of covariates is instead used in subsequent columns as robustness checks, as they do not materially change the estimated treatment effect.

Column (1) presents a baseline DD model with no additional covariates. Column (2) adds raw covariates including gender, marital status, province or territory, income, household size, housing tenure, and education. Column (3) extends the model by including categorical versions of these variables using factor specifications, allowing for more flexible relationships. Column (4) presents our preferred specification, which removes the highest education category due to instability and potential noise in that variable.

Across all model specifications, the estimated DD coefficient (Eligible $\times$ Age 65 or older) is negative and relatively stable, ranging from approximately -0.096 to -0.125. In the preferred specification (Column (4)), the estimate is -0.125 and remains statistically significant at conventional levels.

Since the dependent variable, Y_i , equals 1 if an individual is not in the labour force, the negative coefficient implies that OAS eligibility decreases the likelihood of being out of the labour force. This suggests that OAS eligibility is associated with a lower probability of retirement. The magnitude suggests a decrease of roughly 12.5 percentage points in the probability of being out of the labour force once individuals become eligible for OAS. This is opposite from what we hypothesized using a simple labour-leisure trade-off model, as we expected OAS eligibility to increase retirement rates, but instead find that eligible individuals are less likely to retire.

In addition, the stability of the DD coefficient across specifications suggests that the estimated effect is not highly sensitive to the inclusion of controls. This strengthens the interpretation that the observed change in labour force participation is associated with OAS eligibility rather than compositional differences in observable characteristics.

6 Discussion

A central strength of our approach is that the DD estimator is relatively stable across all specifications. The interaction coefficient remains negative and of similar magnitude as additional controls are introduced, suggesting that the results are not driven by observable differences in demographic and socioeconomic characteristics.

However, a key limitation of this approach is that identification relies on comparing broad age groups rather than individuals at the exact point of eligibility. In particular, our dataset only allows us to compare individuals aged 60-64 and 65-69, rather than observing behaviour precisely before and after turning 65. As a result, we are unable to fully isolate behavioural changes occurring exactly at the OAS eligibility threshold. Ideally, we would observe a finer age distribution that allows for a more precise comparison around age 65, since retirement decisions may respond sharply at the eligibility cutoff rather than gradually across age ranges.

A further limitation is that the DD design relies on the common trends assumption, which requires that absent OAS eligibility, labour force participation would have evolved similarly across the two age groups. This assumption is not directly testable in our setting because the analysis is based on a single cross-section rather than a true panel or repeated time periods.

An additional concern is that the estimated negative effect of OAS eligibility on labour force participation is opposite to the standard prediction from a simple labour-leisure trade-off model, which would suggest that increased non-labour income should reduce labour supply and increase retirement. This discrepancy may indicate that unobserved factors such as health status or labour market attachment may be influencing the results.

These limitations motivate an alternative empirical strategy in future work. In particular, a regression discontinuity design (RDD) around the age 65 cutoff would provide a more precise identification strategy by exploiting the sharp change in eligibility at a single age threshold. This would allow for a cleaner comparison of individuals immediately before and after eligibility, improving causal interpretation. In addition, a placebo test using alternative age cutoffs (such as age 60 or age 70) could be used to test whether similar discontinuities exist where no policy change

occurs. If such effects are not present at placebo cutoffs, this would strengthen confidence that the observed effect at age 65 is driven by OAS eligibility.

7 Conclusion

This paper examines whether Old Age Security (OAS) eligibility affects has a causal effect on labour force participation among Canadians using a difference-in-differences (DD) framework applied to 2021 Canadian Census data. The analysis compares individuals aged 60-64 and 65-69, exploiting the change in eligibility at age 65 to estimate the effect of OAS on retirement behaviour.

Across all specifications, the estimated interaction effect is negative and stable in magnitude. The preferred estimate suggests that OAS eligibility is associated with a 12.5 percentage point decrease in the probability of being out of the labour force. This result implies that OAS eligibility is associated with lower retirement rates and higher labour force participation. This result is contrary to the standard labour-leisure trade-off prediction, which would suggest that additional non-labour income from OAS should increase retirement.

While the results are robust to the inclusion of a rich set of covariates, the analysis is limited by the use of broad age groups rather than exact ages around the eligibility cutoff. This prevents us from precisely isolating behavioural responses at the moment individuals turn 65. In addition, the validity of the DD strategy relies on the common trends assumption, which may be affected by cohort differences between the two age groups.

These limitations suggest that future work could improve upon this analysis by using a regression discontinuity design around the age 65 cutoff, which would allow for a more precise comparison of individuals immediately before and after eligibility. Such an approach would strengthen causal interpretation by reducing concerns about cohort effects and providing a cleaner identification strategy.

Overall, the findings suggest that OAS eligibility is associated with patterns in labour force participation that go against our initial expectations. This highlights the importance of using care-

ful identification strategies when studying the effects of public pension programs on retirement decisions. In future work, having more detailed age data or using alternative methods such as a regression discontinuity design would help better isolate the causal effect of OAS eligibility on labour force participation.

8 References

Ambrozas, D., & Dedewanou, A. (2020). *Canada's retirement income system* (HillStudies, Publication No. 2019-40-E). Library of Parliament.

https://lop.parl.ca/sites/PublicWebsite/default/en_CA/ResearchPublications/201940E

Statistics Canada. (2023). *2021 Census Individuals PUMF User Guide v2.pdf*. 2021 Census Public Use Microdata File (PUMF) Individuals File.

<https://hdl.handle.net/11272.1/AB2/1WTDOP/EV7NLM>

Statistics Canada. (2023). *2021 Census Public Use Microdata File (PUMF) Individuals File*. Abacus Data Network. <https://hdl.handle.net/11272.1/AB2/1WTDOP>

Government of Canada. (2026). *Old Age Security: Do you qualify*. <https://www.canada.ca/en/services/benefits/publicpensions/old-age-security/eligibility.html>